

B.A./B.Sc. (Hons.) SEMESTER V EXAMINATION 2016-17
STATISTICS

STA/STB-501: Applied Statistics

Time: Three hours

Max. Marks: 70

NOTE: Answer any FIVE questions. All questions carry equal marks.

1. Discuss Gross Reproduction Rate (GRR) and Net Reproduction Rate (NRR) in detail with a suitable example. State the demographic interpretation of the exact threshold condition: $NRR = 1$. [cite: 4512]
2. Describe the formulations of the General Fertility Rate (GFR) and the Total Fertility Rate (TFR) with appropriate examples. Provide the exact policy interpretation of the specific numerical statement: $TFR = 2.1$. [cite: 4514, 4515]
3. Define the Crude Death Rate (CDR). Discuss how age-standardization techniques help in overcoming the inherent biological and structural limitations of the crude death rate. [cite: 4517]
4. (a) What are the standard registration and survey methods for obtaining vital statistics? Identify the various administrative sources and practical uses of vital statistics in national planning. [cite: 4519, 4520]
(b) Provide a brief historical account of the Census of India. Who served as the Registrar General and Census Commissioner of India for the historic Census 2011? [cite: 4523]
5. (a) What do you understand by a life table? Write down the structural assumptions of a life table. Describe the mathematical definitions and interrelations of the various columns of a complete life table. [cite: 4527, 4528]
(b) Discuss the major mathematical and empirical curves (such as exponential and logistic functions) used for measuring and forecasting the long-term growth of a population. [cite: 4531]
6. (a) Discuss the structural importance, socioeconomic applications, and statistical limitations of index numbers. [cite: 4533, 4534]
(b) What is a price index number? Explain the precise operational meaning of the current year and the base year in reference to index number formulation. [cite: 4539, 4540]
(c) What is meant by the cost of living index number? Describe step-by-step the statistical procedure followed in its preparation. [cite: 4542, 4543]
7. (a) Discuss the various axiomatic tests of index numbers (Time Reversal, Factor Reversal, and Circular tests) and verify mathematically that Fisher's index number is an ideal index number. [cite: 4546, 4547]
(b) What do you mean by a time series? Detail its critical uses in business planning, economic analysis, and forecasting. [cite: 4549]
(c) Write short notes on the different types of structural components of a time series. Explain the core differences between seasonal variations and cyclical variations using suitable real-world illustrations. [cite: 4555, 4557]
8. (a) Explain in detail the concept of a secular trend. Critically examine and evaluate the relative merits and demerits of various methods used for measuring trends. [cite: 4559]

- (b) What is Statistical Quality Control (SQC)? Discuss the operational need and utility of SQC in modern mass-production industries. Explain why we conventionally consider 3σ control limits in quality control charts. [cite: 4561, 4562]
 - (c) Discuss in detail the statistical construction of the fraction defective (p) chart and the number of defects (c) chart. What is the probabilistic theory behind assuming a Poisson distribution in the case of the c -chart? [cite: 4565, 4566]
 - (d) Discuss the structural construction and cross-interpretation of the mean and range ($\bar{X}, R$) control charts. [cite: 4568]
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External Portals & Academic Resources

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app/>